Order sales are slipping about 10% a year, and 41% of order value never lands

Exploratory Data Analysis | DecodeLabs Data Analytics, Project 2 | Prepared by Favour Daniel

Problem statement

What do 1,200 orders placed between January 2023 and June 2025 tell us about sales patterns, order sizes and where the business is losing value?

Method

I started with the cleaned dataset from Project 1 (no duplicate IDs, no date errors, and 309 blank coupon codes marked "No Coupon"). I then worked out the count, mean and median for each number column, checked for outliers with the IQR method, compared sales over time, and broke sales down by product, referral source, coupon, payment method and order status. Correlation was used to see which factors move with order value. The data ends in June 2025, so 2025 is a half year and is compared with the same months of earlier years.

Key findings

1. **Sales are falling.** Comparing January to June in each year, sales dropped from about 286,500 in 2023 to 257,100 in 2024 and 231,900 in 2025, roughly 10% a year. The best month was June 2024 (about 68,100) and the weakest was April 2023 (about 27,800). The first half of the year is consistently stronger, and September is the weakest month overall.
2. **The typical order is smaller than the average.** The median order is 824, against a mean of 1,054, because a few large orders pull the average up. Eight orders sit above the outlier limit of about 3,330. All are five-item purchases at high unit prices, and they pass the checks against quantity × price and cart size, so I kept them as genuine large orders.
3. **Four in ten orders are cancelled or returned.** Only 231 orders (19%) were delivered. Cancelled and returned orders make up 497 orders and about 519,700 of the 1,264,762 total, so the headline sales figure is the value of orders placed, not money kept.
4. **Segments are close, with a few gaps.** Chair and Printer lead at about 195,600 each, and Phone is lowest at 151,700, about 22% below. Instagram is the strongest referral source (275,285) and Referral the weakest (226,816). Credit Card orders are the largest on average (about 1,128, against 1,002 for Debit Card).
5. **Coupons do not visibly lift order size.** The average order is about 1,070 with FREESHIP and 1,043 with no coupon.
6. **Higher prices and bigger quantities both raise order value.** Unit price (r = 0.72) and quantity (r = 0.62) both rise with order value, while the number of items in the cart has only a loose link (r = 0.39). Price and quantity are unrelated to each other (r = 0.02). Correlation shows a link, not a cause.

Recommendations

1. Find out why 41% of orders are cancelled or returned. It is the biggest single leak in the data.
2. Investigate the yearly decline, and plan promotions for July to December, when sales are weakest.
3. Put more marketing into Instagram, and review whether the Referral channel is worth the effort.
4. Look at why Phone sells least, whether it is price, stock or how it is listed.
5. Test whether coupons are worth the discount, since they do not raise order size.

Note: the differences between most segments are small, so they are best read as slight leads rather than clear wins.